

# HTTP/3 is coming

The web is built on HTTP. There is no question about it. It's pretty amazing how well this protocol fared. Here is a quick recap: in 1991, Tim-Berneres-Lee proposes HTTP 0.9 to support his idea for a World Wide Web. In 1996, The HTTP Working Group publishes HTTP 1.0 as the informational RFC 1945 to enable the internet boom of the late 1990s. In 1997, the first official RFC 2068 for HTTP 1.1 is published. In 1999, RFC 2616 adds a number of improvements to HTTP 1.1 and remains the dominant standard for two decades. In 2015, HTTP/2 is published, based on the SPDY protocol by Google, and all major browsers add support for it.

gRPC is built on top of HTTP/2, which fixes a lot of issues with previous revisions of the HTTP and provides the following features:

- Binary framing and compression
- Multiplexing using streams (multiple requests on the same TCP connection)
- Better flow control
- Server push

That sounds great. What will HTTP/3 give us? It offers the same feature set of HTTP/2. However, HTTP/2 is based on TCP, which doesn't offer streams. That means that streams are implemented at the HTTP/2 level. HTTP/3 is based on QUIC, a reliable transport over UDP. The details are out of scope, but the bottom line is that HTTP/3 will have much better performance and is always secure.

It may still take a while for broad HTTP/3 adoption, because many enterprises block or rate limit UDP on their networks. However, the benefits are compelling and gRPC over HTTP/3 will have even a bigger edge in performance compared to REST APIs.

Those are the primary future trends that will impact microservices. Let's see what is next for Kubernetes.

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